

Clinical Correlates of Episodicity in Juvenile Mania

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ABSTRACT

Objective: Researchers debate whether the diagnostic criteria for mania should differ between children and adults. Specifically, although the *Diagnostic and Statistical Manual of Mental Disorders* (fourth edition; DSM-IV) requires episodic mood changes, children commonly are diagnosed as manic on the basis of chronic irritability. In this preliminary study, children carrying a diagnosis of bipolar disorder (BPD) in the community were classified as having either episodic or chronic symptoms. We hypothesized that the episodic group would be more likely to have a history of psychosis and a parental history of BPD, whereas the chronic group would be more likely to have conduct disorder.

Methods: Parents of children carrying the BPD diagnosis were interviewed on the telephone to obtain psychiatric and family histories. Children were considered episodic ($n = 34$) if they had a history of one or more DSM-IV manic/hypomanic episodes meeting full duration criteria and chronic ($n = 53$) if they had no discernable episodes.

Results: The episodic group was more likely to have had psychosis, parental history of BPD, and to have experienced each manic symptom except for irritability and psychomotor agitation. Children in the episodic group were also more likely to have had a depressive episode meeting full DSM-IV criteria and were more likely to have made a suicide attempt. Children in the chronic group were not more likely to meet criteria for conduct disorder but were more likely to exhibit violence toward others.

Conclusions: These preliminary data indicate that, among children being treated for BPD in the community, those with discrete episodes of mania may be more likely to have a lifetime history of psychosis and a parental history of BPD. The latter hypothesis should be tested in a sample where relatives are interviewed directly.

INTRODUCTION

DIAGNOSTIC CRITERIA for juvenile mania have been debated in the literature (Biederman et al. 1998; Carlson 1998; Geller et al. 1998). The importance of discernable manic episodes, and their duration, is an important part of this

debate (Biederman et al. 1998; Leibenluft et al. 2003). Clinical researchers have observed that children with bipolar disorder (BPD) may have more frequent and briefer manic episodes than do adults with the illness (Biederman et al. 1998; Carlson 1998; Geller et al. 1998). Because of this developmental differ-

ence in clinical presentation, some researchers have suggested that the *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition (DSM-IV; American Psychiatric Association 1994), criteria for mania should be modified for children by eliminating the requirement for discrete episodes and/or by shortening the required duration of episodes (currently 7 days for mania and 4 days for hypomania) (Biederman et al. 1998).

The question of episodes and their duration is important in the differential diagnosis of BPD versus attention deficit hyperactivity disorder (ADHD). Although there is some overlap in the diagnostic criteria for ADHD and mania, one difference between the illnesses is that the mania of BPD is episodic, whereas the symptoms of ADHD are not. DSM-IV operationalizes episodicity in mania by stating that, in order to meet criteria for the illness, one must have a distinct change of mood (elevated, expansive, or irritable), lasting at least 7 days for mania and 4 days for hypomania (criterion "A" of the DSM-IV criteria), and occurring simultaneously with a number of other symptoms (i.e., criterion "B": distractibility, pressured speech, decreased need for sleep, etc.) (American Psychiatric Association 1994).

Another question regarding the diagnosis of BPD in children concerns whether euphoria and/or grandiosity, rather than the nonspecific symptom of irritability, should be required to meet criteria for mania (Geller et al. 1998, 2002). The literature on adult BPD clearly indicates that mania can be associated with irritability rather than euphoria, although discernable episodes are still described (Cassidy et al. 1998, 2001). If neither episodes nor euphoria or grandiosity are required to meet criteria for mania, it can be difficult to distinguish mania from other childhood-onset illnesses characterized by chronic irritability, including ADHD with irritability, oppositional defiant disorder, and various admixtures of depressive and anxiety disorders.

We were interested in exploring how those children diagnosed with BPD in the community who had discrete episodes of illness differed from those who were similarly diagnosed but whose symptoms took a chronic course. We addressed this question using pilot data ob-

tained while screening children for a longitudinal study of the pathophysiology of childhood-onset BPD. All of the children who were screened for the study had been diagnosed as having BPD by clinicians in the community. However, data indicate that only 40% of children who receive clinical diagnoses of BPD also meet research criteria (Pogge et al. 2001). Indeed, our early experience recruiting for this study indicated that many children carrying the clinical diagnosis of BPD did not, as required by DSM-IV, have distinct episodes of mania. This afforded us the opportunity to compare, in terms of clinical presentation and family history, those children diagnosed in the community with BPD who experience distinct episodes of mania with children who carry the same diagnosis but instead experience chronic, nonepisodic symptoms.

The organizing hypothesis for this study was that, compared with children with chronic symptoms, those with episodic illness would more closely resemble the adult phenotype of BPD. In addition to the presence of episodes, two important clinical features of the adult phenotype are psychosis and a positive family history of BPD (Potash et al. 2003; Rice et al. 1987). Therefore, we hypothesized that children who had experienced at least one discrete manic/hypomanic episode meeting full DSM-IV duration criteria (EPI group) would differ from those with a chronic course (CHR group) in that the former would be more likely to have a lifetime history of psychosis and a parental history of BPD. Conversely, we hypothesized that the CHR group would be more likely to meet criteria for conduct disorder, because both groups exhibit chronic symptoms of aggression, property destruction, and/or rule violation (Biederman et al. 1999). In addition, published data indicate that children who receive the clinical diagnosis of BPD but who do not meet research criteria often meet criteria for conduct disorder (Pogge et al. 2001). Finally, we performed preliminary analyses on the association between episodicity and the presence of cardinal symptoms (i.e., symptoms of mania that are relatively specific to the illness and are rarely seen in other axis I disorders, such as elation or grandiosity/increased self-esteem) (Geller et al. 1998, 2002). We hypothesized that

children in the EPI group would be more likely to exhibit cardinal manic symptoms than those in the CHR group.

METHODS

Subjects for this study were obtained through nationwide recruitment for a longitudinal study on the phenomenology and pathophysiology of BPD in children ages 6–17 years. The recruitment was accomplished through Web-based advertisements and mailings to physicians, psychologists, pediatricians, and school counselors. Subjects were included in this pilot study if (1) their parents expressed interest in enrolling the child in the study, (2) the child was currently in treatment for BPD with a psychiatrist in the community, and (3) the parent was able to complete the telephone screening interview. In addition, parents signed written consent forms, and children signed assent forms (these forms were mailed to the subjects, signed, and mailed back). The parents understood that agreeing to participate in the phone screen study did not guarantee acceptance to the on-site longitudinal study. Both the on-site investigation and this pilot study were approved by the National Institute of Mental Health Investigational Review Board.

As part of the screening for the longitudinal study, the parents underwent a 2- to 3-hour semistructured telephone interview by clinicians blind to the hypotheses of this pilot analysis. Mental health professionals (master's level or above), trained to reliability on the Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children–Present and Lifetime version (Kaufman et al. 1997), conducted the interviews. The telephone interview was modeled on the Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children and reviewed all childhood psychiatric disorders. Medical records were also reviewed to confirm parental report of episodic (vs. chronic) symptoms, presence of criterion B symptoms, comorbid diagnoses, and symptoms of psychosis. The child was considered to have a manic/hypomanic episode if he or she had experienced a 7-day (or 4-day, in the case of hypomania) history of ei-

ther euphoria or irritability. If the mood disturbance was primarily irritability, he or she had to show four symptoms of criterion B. If euphoria was present, he or she had to show three symptoms of criterion B. Those criterion B symptoms grouped together in single items in the DSM-IV (i.e., grandiosity/increased self-esteem, increased goal-directed activity/psychomotor agitation, and racing thoughts/flight of ideas) were not separated when the determination was made as to whether an episode met DSM-IV criteria or when the number of criterion B symptoms was reported (i.e., the number of criterion B symptoms is the number of DSM-IV B symptoms). However, these grouped symptoms were separated subsequently in order to examine the frequency of each individual symptom. This was done because they were clinically distinguishable and because, in some instances, criteria that are combined into one DSM-IV criterion differ in the extent to which they are relatively unique to mania (e.g., increase in goal-directed activity vs. psychomotor agitation).

The mood disturbance was considered episodic if the parents felt it was a distinct change from the child's baseline. For example, the number of hours slept each night could be different, or the parents might note a marked change in activity level, in the amount of homework that the child could do, and so forth. Most children were impaired at baseline, but the mood change and the associated criterion B symptoms (which had to occur at the same time as the mood change) together had to cause significantly more impairment than was present at baseline. With regard to psychosis, the child was considered to have experienced psychotic symptoms if he or she was noted to have had hallucinations, delusions, or evidence of a disorder of thought processing. Equivocal instances (e.g., one instance of hearing a voice calling his or her name) were discounted. Parental history of psychiatric disorder was considered positive only if the parent had been diagnosed with the disorder by a psychiatrist (the interviewed parent was also asked about the other parent). All available information was used to reach a best-estimate consensus among the research team as to whether the child had a lifetime history of at least one full duration episode of mania/hypomania (i.e., a dis-

tinct period that could be differentiated from baseline). The members of the research team included the clinicians performing the telephone interviews as well as two board-certified child psychiatrists and one board-certified adult psychiatrist, all of whom are experienced in the assessment of childhood-onset BPD.

Of the first 153 subjects, written consents and assents were received for 125. Ten children who had episodes lasting fewer than 4 days were excluded from the sample because they did not fit clearly into either the EPI or the CHR group. An additional 28 children were not included: 7 because of information deemed unreliable by the interviewer; 7 because of medical conditions that might have caused their symptoms; 6 because the parent failed to complete the interview; 6 because mood symptoms appeared to be secondary to medication effects; 1 because of significant pervasive developmental disorder symptoms; and 1 because of concomitant substance abuse. Thus, the data presented here are from 87 children.

Of the 87 children in the sample, 34 met criteria for at least one lifetime manic episode meeting DSM-IV criteria. These 34 children comprised the EPI group. The remaining 53 children had chronic symptoms (CHR group) without discernable episodes. All children with a chronic course were included in the CHR group, regardless of the number of criterion B symptoms.

Consistent with our hypotheses, we compared the EPI and CHR groups in terms of parental history of BPD, child lifetime history of psychotic symptoms, and presence of conduct disorder. We also performed exploratory analyses comparing the prevalence of each of the manic and depressive symptoms in the two groups and examining the relation between cardinal symptoms (elation or grandiosity) and episodicity within the sample as a whole. The data were analyzed using the Statistical Package for the Social Sciences using chi-square analyses for categorical data and *t* tests for continuous variables (*t* tests were adjusted if unequal variances were found). Because of the exploratory nature of the study, significance was set at $p \leq 0.05$. Bonferroni correction was used for the analysis of the frequency of each of the manic and depressive symptoms. To ex-

plore the association between cardinal symptoms (i.e., symptoms relatively specific for mania; in this study, we used elation and grandiosity) and episodicity, we used a Mann-Whitney *U* test, because the number of cardinal symptoms was not normally distributed.

RESULTS

As shown in Table 1, the two groups did not differ in age (range 6–17 years; mean EPI 11.12 ± 2.66 years, mean CHR 11.09 ± 2.54 years), age of first psychiatric/psychological evaluation (mean EPI 5.97 ± 2.76 years, mean CHR 6.41 ± 2.60 years), duration of illness (mean EPI 5.06 ± 2.12 years, mean CHR 4.73 ± 2.41 years), gender (EPI 68% male, CHR 64% male), number of psychotropic medications (mean EPI 3.29 ± 1.38 , mean CHR 3.45 ± 1.84), and number of psychiatric hospitalizations (mean EPI: 1.59 ± 2.23 , mean CHR 1.17 ± 1.22) (all comparisons not significant; see Table 1). The two groups also did not differ in comorbid disruptive behavior disorders (ADHD: EPI 77%, CHR 67%; oppositional defiant disorder: EPI 65%, CHR 81%; conduct disorder: EPI 22%, CHR 33%; all comparisons not significant; see Table 1). Although our hypothesis regarding comorbidity with conduct disorder was not supported, more children in the CHR group exhibited violent behavior toward others (CHR 49%, EPI 24%; $\chi^2 = 5.67$, $df = 1$, $p = 0.024$).

Children in the EPI group were more likely to have experienced psychotic symptoms (EPI 62%, CHR 38%; $\chi^2 = 4.80$, $df = 1$, $p = 0.034$) and to have a parent who had been given a diagnosis of BPD by a psychiatrist (EPI 29%, CHR 9%, $\chi^2 = 5.79$, $df = 1$, $p = 0.018$). The groups did not differ in parental history of other psychiatric illnesses. All of the DSM-IV criteria for mania, except irritability and psychomotor agitation, occurred more often in the EPI group; irritability and psychomotor agitation occurred with comparable frequency in the two groups (Table 2). These findings remained significant after adjustments were made for multiple comparisons. In addition, the EPI group experienced more of the B mania criteria (mean number of B criteria: EPI 5.91 ± 1.14 , CHR 2.32 ± 1.59 ; $t = 12.26$, $df = 83.91$, $p < 0.0001$).

TABLE 1. DEMOGRAPHIC INFORMATION

	<i>Episodic</i> (n = 34)	<i>Chronic</i> (n = 53)
Age (mean years ± SD)	11.12 ± 2.66	11.09 ± 2.54
Age of first psychiatric/psychological evaluation (mean years ± SD)	5.97 ± 2.76	6.41 ± 2.60
Duration of illness (mean years ± SD)	5.06 ± 2.12	4.73 ± 2.41
Gender (% male)	68%	64%
Number of medications (mean ± SD)	3.29 ± 1.38	3.45 ± 1.84
Number of psychiatric hospitalizations (mean ± SD)	1.59 ± 2.23	1.17 ± 1.22
% with comorbid ADHD	77% (n = 26)	67% (n = 46)
% with comorbid ODD	65% (n = 26)	81% (n = 42)
% with comorbid conduct disorder	22% (n = 27)	33% (n = 45)

Note: There were no significant differences between the two groups on any of the above measures. ADHD = attention deficit hyperactivity disorder; ODD = oppositional defiant disorder.

When parents were asked whether their child ever experienced each of the individual depressive symptoms, the percentage having experienced each symptom did not differ significantly between groups. However, when the duration and number of symptoms occurring together were considered, as required by DSM-IV, a significantly greater percentage of children in the EPI group met full DSM-IV criteria (i.e., five or more DSM-IV symptoms over a 2-week period) for at least one lifetime episode of major depressive disorder (EPI 70%, CHR 45%; $\chi^2 = 4.196, df = 1, p = 0.049$). In con-

trast, more children in the CHR group had their longest depressive episodes lasting a day or less (percentage of children with their longest depressive episode lasting a day or less: EPI 10%, CHR 30%, $\chi^2 = 4.23, df = 1, p = 0.048$). Although the groups did not differ in terms of the prevalence of a lifetime history of suicidal thoughts, children in the EPI group were significantly more likely to have made a suicide attempt (EPI 47%, CHR 15%; $\chi^2 = 10.59, df = 1, p = 0.001$). Among children in the CHR group, 42 (79%) had no cardinal symptoms (grandiosity/in-

TABLE 2. MANIC SYMPTOMS

<i>Manic symptoms</i>	<i>Episodic</i> (n = 34)	<i>Chronic</i> (n = 53)	χ^2	<i>p</i>
Elation	30 (88%)	5 (9%)	53.49	<0.0001
Increased goal-directed activity	29 (85%)	7 (13%)	44.37	<0.0001
Grandiosity	23 (68%)	4 (7.5%)	34.96	<0.0001
Decreased need for sleep	26 (76%)	9 (17%)	30.49	<0.0001
Racing thoughts	25 (74%)	10 (19%)	25.74	<0.0001
Excessive pleasure seeking	22 (65%)	8 (15%)	22.57	<0.0001
Increased self-esteem	21 (62%)	8 (15%)	20.30	<0.0001
Pressured speech	30 (88%)	21 (40%)	20.18	<0.0001
Distractibility	30 (88%)	27 (51%)	12.75	<0.0001
Flight of ideas	16 (47%)	7 (13%)	12.20	0.001
Irritability	28 (82%)	41 (77%)	0.32	ns
Psychomotor agitation	24 (71%)	39 (74%)	0.09	ns

Data in the Episodic and Chronic columns are expressed as *n* (%). ns = not significant.

creased self-esteem, elation); 8 (15%) had one, and 3 (6%) had two. Among the EPI group, 0 had no cardinal symptoms, 7 (21%) had one, and 27 (79%) had two. The distribution was significantly different for the two groups, with a Mann-Whitney *U* test value of 89.5 ($p < 0.0001$).

DISCUSSION

Recent controversy about the diagnosis of juvenile mania has focused on the nature and course of the “abnormal mood.” The DSM-IV criteria require a distinct period of abnormally and persistently elevated, expansive, or irritable mood, with duration lasting 4 days for hypomania and 7 days for mania. Some investigators have argued that mania in prepubertal children instead presents as a chronic condition without clearly discernable episodes (Biederman et al. 1998; Wozniak et al. 1995). Thus, in diagnosing mania in this age group, an important issue becomes whether episodic mood changes should be required and, if so, what the required duration should be.

To shed some light on this question, we compared the clinical characteristics and family history of children diagnosed with BPD in

the community who had at least one manic episode meeting full DSM-IV duration criteria with a sample of children who had been diagnosed with BPD by their treating psychiatrists but had no discernable episodes. We hypothesized that the children with a history of at least one clear manic episode would have clinical features consistent with the presentation of adult mania (i.e., history of psychosis and a parental history of BPD). These hypotheses were confirmed.

We found that cardinal symptoms of mania and episodicity are strongly associated. Due to the small sample size, it is difficult to determine which of the two factors is more closely associated with psychosis and parental history of BPD. Multivariate analyses with a larger sample are needed to answer this question. Currently, it appears that both the presence of cardinal symptoms and episode duration are important in the assessment of pediatric mania.

Of importance, both groups of children (CHR and EPI) were severely impaired, as evidenced by the mean number of psychiatric medications and hospitalizations. Psychosis, depressive episodes meeting full DSM-IV criteria, and suicide attempts were all more common in the EPI group, placing their clinical

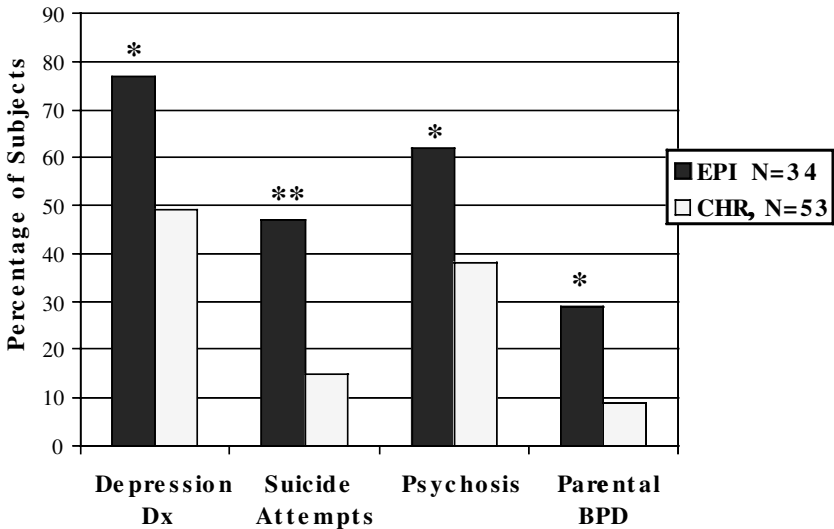


FIG. 1. Associated symptoms. “Depression dx” refers to lifetime history of an episode meeting *Diagnostic and Statistical Manual of Mental Disorders* (fourth edition) criteria for major depressive disorder. Similarly, “suicide attempts” and “psychosis” refer to the percentage of subjects in each group who have a lifetime history of that feature. “Parental BPD” refers to the percentage in each group who have a parent who has been diagnosed by a psychiatrist as having bipolar disorder. CHR = the group showing chronic symptoms; EPI = the group showing episodic symptoms. * $p < 0.05$. ** $p = 0.001$.

phenotype on a continuum with that of adult BPD (see Fig. 1). Furthermore, the EPI group was more likely to have experienced symptoms that are not typically seen in other axis I diagnoses (the cardinal symptoms of elation and grandiosity) (Geller et al. 1998). Other, less specific symptoms (irritability and psychomotor agitation) were present equally in both groups. Thus, it appears that manic symptoms, clear depressive episodes, psychosis, and history of suicide attempts differentiated the two groups, and in each instance the EPI group more closely resembled what would be expected in a sample of adults with BPD.

In addition, these preliminary data appear to show that the children in the EPI group are more likely to have a parent diagnosed with BPD. However, the fact that parents (usually mothers) were asked about other family members in an unstandardized way constitutes a significant methodological limitation of the study. Our data indicate that a worthwhile subsequent study would involve systematic interviewing of relatives of children with episodic and chronic presentations, by interviewers blind to the clinical presentation of the proband. In addition, Hirschfeld et al. (2003) noted that adult patients with BPD often remain undiagnosed for a significant time, and it is possible that parents of children in the CHR group are more likely than parents of children in the EPI group to have a form of BPD that is less likely to be recognized and diagnosed.

Another significant limitation of our study is that, because we did not perform in-person standardized research diagnostic interviews with both parents and children, we cannot definitively state whether or not the children would meet research diagnostic criteria for BPD. However, with regard to the CHR group, the fact that, by definition, they would not meet the A criterion in DSM-IV, combined with their low rate of B symptoms (mean 2.32 ± 1.59), makes it unlikely that they would meet research diagnostic criteria. In that sense, our study can be seen as a replication of Pogge et al.'s (2001) work, which found a poor concordance between community diagnoses of mania and rigorous research screening.

As noted above, other limitations of the study include the fact that information was ob-

tained by parental report only, and by telephone; however, telephone interviews have been used to diagnose BPD in adults (Hirschfeld et al, 2003). Ideally, clinician-administered, standardized interviews should have been conducted with both children and parents, but limited resources precluded this. Also, information regarding mixed episodes (as diagnosed by DSM-IV criteria) was difficult to obtain reliably in retrospect, because parents were unclear as to the precise temporal relation of the depressive and the manic symptoms and the extent to which they overlapped or were sequential; this is another limitation of the study. Results from this study should be taken as preliminary and require replication with a larger sample size and on-site evaluation.

In sum, these preliminary data lend some support to the suggestion that, at this time, the narrow phenotype of juvenile mania (Geller et al. 2002) might be reserved for children exhibiting clear, full-duration manic episodes, possibly with cardinal symptoms. The nosological status of children with chronic irritability, depressive symptoms, and disruptive behavior remains unclear and requires further study (Leibenluft et al. 2003). Although the children in the CHR group were not more likely to meet criteria for conduct disorder, as hypothesized, they did exhibit more violence toward others. This supports the need for further research on the relation between chronic mood symptoms and behavioral disorders (Pine et al. 1998). Indeed, the fact that the children in the CHR group are as impaired as those in the EPI group highlights the importance of further research on the CHR group. At present, there appears to be preliminary evidence that, in children with significant and impairing mood symptoms, a history of at least one full-duration hypomanic/manic episode may be more likely to be associated with psychosis, a major depressive episode, a suicide attempt, and possibly a family history of BPD.

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